

# Corey Oses

Materials Science and Engineering, Johns Hopkins University

[Work Experience](#) · [Education](#) · [Patents](#) · [Journal Publications](#) · [Book Publications](#) ·  
[Teaching Experience](#) · [Service](#) · [Press and News Releases](#) · [Honors and Awards](#)

email [corey@jhu.edu](mailto:corey@jhu.edu)  
phone (W) +1 (410) 516 5779  
website [coreyoses.com](http://coreyoses.com)  
Google Scholar [user=Za7m4CMAAAAJ](https://scholar.google.com/citations?user=Za7m4CMAAAAJ) · citations: 9991 (YTD 1163) · h-index: 36

## Work Experience

|                     |              |                                                                        |
|---------------------|--------------|------------------------------------------------------------------------|
| Assistant Professor | 2022–present | Johns Hopkins University                                               |
| Postdoctoral Fellow | 2018–2022    | Duke University                                                        |
| Internship          | Summer 2013  | Cornell High Energy Synchrotron Source (BioSAXS on F2 and G Beamlines) |
| Internship          | Summer 2012  | Cornell High Energy Synchrotron Source (Capillary Optics Group)        |

## Education

|       |           |                                                                                                                                                                                                                                                                                                                        |
|-------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ph.D. | 2013–2018 | Duke University<br><b>Department:</b> Mechanical Engineering and Materials Science<br><b>Thesis:</b> <i>Machine learning, phase stability, and disorder with the Automatic Flow Framework for Materials Discovery</i><br><b>DukeSpace:</b> <a href="https://hdl.handle.net/10161/18254">hdl.handle.net/10161/18254</a> |
| B.Sc. | 2009–2013 | Cornell University<br><b>Department:</b> Applied and Engineering Physics<br><b>Thesis:</b> <i>Plume Propagation Simulation for Pulsed Laser Deposition</i>                                                                                                                                                             |

## Journal Publications<sup>†</sup> contributed equally \* corresponding

2026

49. X. Xu, G. Han, T. Li, J. Hu, G. Qiu, S.-Y. Tseng, G. H. Han, A. Dupuy, A. S. Hall & C. Oses\*, *Stabilizing Iodine in Pyrochlore: Toward New Nuclear Waste Forms*, *Precis. Chem.* (2026). DOI: [10.1021/prechem.5c00467](https://doi.org/10.1021/prechem.5c00467). [PDF]

2025

48. G. Han<sup>†</sup>, T. Li<sup>†</sup>, X. Xu, J. Lee, S. Sequeira, A. Ajith & C. Oses\*, *The search for high-entropy fuel-cell catalysts using disorder descriptors*, *Nano Futures* **9**, 045001 (2025). DOI: [10.1088/2399-1984/ae19b0](https://doi.org/10.1088/2399-1984/ae19b0). [PDF]  
47. C. Oses\*, T. Li, X. Xu, G. Han, G. Qiu & J. R. Owens, *Beyond the four core effects: revisiting thermoelectrics with a high-entropy design*, *Mater. Horiz.* **12**, 5946–5956 (2025). DOI: [10.1039/D5MH00356C](https://doi.org/10.1039/D5MH00356C). [PDF]  
• This paper was selected to feature in **2025 Most Popular Articles** by the Royal Society of Chemistry: *Materials Horizons* (2026).  
46. G. Qiu, T. Li, X. Xu, Y. Liu, M. Niyogi, K. Cariaga & C. Oses\*, *High entropy powering green energy: hydrogen, batteries, electronics, and catalysis*, *npj Comput. Mater.* **11**, 145 (2025). DOI: [10.1038/s41524-025-01594-6](https://doi.org/10.1038/s41524-025-01594-6). [PDF]

2024

45. T. Gong, G. Qiu, M.-R. He, O. V. Safonova, W.-C. Yang, D. Raciti, C. Oses & A. S. Hall, *Atomic Ordering-Induced Ensemble Variation in Alloys Governs Electrocatalyst On/Off States*, *J. Am. Chem. Soc.* **147**(1), 510–518 (2024). DOI: [10.1021/jacs.4c11753](https://doi.org/10.1021/jacs.4c11753). [PDF]  
44. K. S. Vecchio, S. Curtarolo, K. Kaufmann, T. J. Harrington, C. Oses & C. Toher, *Fermi energy engineering of enhanced plasticity in high-entropy carbides*, *Acta Mater.* **276**, 120117 (2024). DOI: [10.1016/j.actamat.2024.120117](https://doi.org/10.1016/j.actamat.2024.120117). [PDF]

43. M. L. Evans, J. Bergsma, A. Merkys, C. W. Andersen, O. B. Andersson, D. Beltrán, E. Blokhin, T. M. Boland, R. Castañeda Balderas, K. Choudhary, A. Díaz Díaz, R. Domínguez García, H. Eckert, K. Eimre, M. E. Fuentes-Montero, A. M. Krajewski, J. J. Mortensen, J. M. Nápoles-Duarte, J. Pietryga, J. Qi, F. d. J. Trejo Carrillo, A. Vaitkus, J. Yu, A. C. Zettel, P. Baptista de Castro, J. Carlsson, T. F. T. Cerqueira, S. Divilov, H. Hajiyani, F. Hanke, K. Jose, **C. Oses**, J. Riebesell, J. Schmidt, D. Winston, C. Xie, X. Yang, S. Bonella, S. Botti, S. Curtarolo, C. Draxl, L. E. Fuentes Cobas, A. Hospital, Z.-K. Liu, M. A. L. Marques, N. Marzari, A. J. Morris, S. P. Ong, M. Orozco, K. A. Persson, K. S. Thygesen, C. Wolverton, M. Scheidgen, C. Toher, G. J. Conduit, G. Pizzi, S. Gražulis, G.-M. Rignanese & R. Armiento, *Developments and applications of the OPTIMADE API for materials discovery, design, and data exchange*, *Digit. Discov.* **3**, 1509–1533 (2024). DOI: [10.1039/D4DD00039K](https://doi.org/10.1039/D4DD00039K). [PDF]
42. S. Divilov<sup>†</sup>, H. Eckert<sup>†</sup>, D. Hicks<sup>†</sup>, **C. Oses**<sup>†</sup>, C. Toher<sup>†</sup>, R. Friedrich, M. Esters, M. J. Mehl, A. C. Zettel, Y. Lederer, E. Zurek, J.-P. Maria, D. W. Brenner, X. Campilongo, S. Filipović, W. G. Fahrenholtz, C. J. Ryan, C. M. DeSalle, R. J. Crealese, D. E. Wolfe, A. Calzolari & S. Curtarolo, *Disordered enthalpy-entropy descriptor for high-entropy ceramics discovery*, *Nature* **625**, 66–73 (2024). DOI: [10.1038/s41586-023-06786-y](https://doi.org/10.1038/s41586-023-06786-y). [PDF]
41. A. B. Peters, D. Zhang, S. Chen, C. Ott, **C. Oses**, S. Curtarolo, I. McCue, T. Pollock & S. E. Prameela, *Materials Design for Hypersonics*, *Nat. Commun.* **15**, 3328 (2024). DOI: [10.1038/s41467-024-46753-3](https://doi.org/10.1038/s41467-024-46753-3). [PDF]
  - This paper was selected for **Editors' Highlight** by Springer Nature (2024).

## 2023

40. D. E. Wolfe, C. M. DeSalle, C. J. Ryan, R. E. Slapikas, R. T. Sweny, R. J. Crealese, P. A. Kolonin, S. P. Stepanoff, A. Haque, S. Divilov, H. Eckert, **C. Oses**, M. Esters, D. W. Brenner, W. G. Fahrenholtz, J.-P. Maria, C. Toher, E. Zurek & S. Curtarolo, *Influence of Processing on the Microstructural Evolution and Multiscale Hardness in Titanium Carbonitrides (TiCN) Produced via Field Assisted Sintering Technology*, *Materialia* **27**, 101682 (2023). DOI: [10.1016/j.mtl.2023.101682](https://doi.org/10.1016/j.mtl.2023.101682). [PDF]
39. **C. Oses**, M. Esters, D. Hicks, S. Divilov, H. Eckert, R. Friedrich, M. J. Mehl, A. Smolyanyuk, X. Campilongo, A. van de Walle, J. Schroers, A. G. Kusne, I. Takeuchi, E. Zurek, M. Buongiorno Nardelli, M. Fornari, Y. Lederer, O. Levy, C. Toher & S. Curtarolo, *aflow++: a C++ framework for autonomous materials design*, *Comput. Mater. Sci.* **217**, 111889 (2023). DOI: [10.1016/j.commatsci.2022.111889](https://doi.org/10.1016/j.commatsci.2022.111889). [PDF]
  - This paper was selected for **Editor's Choice** by Elsevier (2022).
38. M. Esters, **C. Oses**, S. Divilov, H. Eckert, R. Friedrich, D. Hicks, M. J. Mehl, F. Rose, A. Smolyanyuk, A. Calzolari, X. Campilongo, C. Toher & S. Curtarolo, *aflow.org: a web ecosystem of databases, software and tools*, *Comput. Mater. Sci.* **216**, 111808 (2023). DOI: [10.1016/j.commatsci.2022.111808](https://doi.org/10.1016/j.commatsci.2022.111808). [PDF]
37. M. Esters<sup>†</sup>, A. Smolyanyuk<sup>†</sup>, **C. Oses**, D. Hicks, S. Divilov, H. Eckert, X. Campilongo, C. Toher & S. Curtarolo, *QH-POCC: taming tiling entropy in thermal expansion calculations of disordered materials*, *Acta Mater.* **245**, 118594 (2023). DOI: [10.1016/j.actamat.2022.118594](https://doi.org/10.1016/j.actamat.2022.118594). [PDF]

## 2022

36. A. Calzolari, **C. Oses**, C. Toher, M. Esters, X. Campilongo, S. P. Stepanoff, D. E. Wolfe & S. Curtarolo, *Plasmonic high-entropy carbides*, *Nat. Commun.* **13**, 5993 (2022). DOI: [10.1038/s41467-022-33497-1](https://doi.org/10.1038/s41467-022-33497-1). [PDF]
35. X. Wang, D. M. Proserpio, **C. Oses**, C. Toher, S. Curtarolo & E. Zurek, *The Microscopic Diamond Anvil Cell: Stabilization of Superhard, Superconducting Carbon Allotropes at Ambient Pressure*, *Angew. Chem.* **61**(32), e202205129 (2022). DOI: [10.1002/anie.202205129](https://doi.org/10.1002/anie.202205129). [PDF]
34. H. J. Kulik, T. Hammerschmidt, J. Schmidt, S. Botti, M. A. L. Marques, M. Boley, M. Scheffler, M. Todorović, P. Rinke, **C. Oses**, A. Smolyanyuk, S. Curtarolo, A. Tkatchenko, A. P. Bartók, S. Manzhos, M. Ihara, T. Carrington, J. Behler, O. Isayev, M. Veit, A. Grisafi, J. Nigam, M. Ceriotti, K. T. Schütt, J. Westermayr, M. Gastegger, R. J. Maurer, B. Kalita, K. Burke, R. Nagai, R. Akashi, O. Sugino, J. Hermann, F. Noé, S. Pilati, C. Draxl, M. Kuban, S. Rigamonti, M. Scheidgen, M. Esters, D. Hicks, C. Toher, P. V. Balachandran, I. Tamblyn, S. Whitelam, C. Bellinger & L. M. Ghiringhelli, *Roadmap on Machine Learning in Electronic Structure*, *Electron. Struct.* **4**(2), 023004 (2022). DOI: [10.1088/2516-1075/ac572f](https://doi.org/10.1088/2516-1075/ac572f). [PDF]
33. A. G. Kusne, A. McDannald, B. DeCost, **C. Oses**, C. Toher, S. Curtarolo, A. Mehta & I. Takeuchi, *Physics in the Machine: Integrating Physical Knowledge in Autonomous Phase-Mapping*, *Front. Phys.* **10**, 815863 (2022). DOI: [10.3389/fphy.2022.815863](https://doi.org/10.3389/fphy.2022.815863). [PDF]
32. C. Toher, **C. Oses**, M. Esters, D. Hicks, G. N. Kotsonis, C. M. Rost, D. W. Brenner, J.-P. Maria & S. Curtarolo, *High-entropy ceramics: Propelling applications through disorder*, *MRS Bull.* **47**, 194–202 (2022). DOI: [10.1557/s43577-022-00281-x](https://doi.org/10.1557/s43577-022-00281-x). [PDF]

## 2021

31. M. Esters, **C. Oses**, D. Hicks, M. J. Mehl, M. Jahnátek, M. D. Hossain, J.-P. Maria, D. W. Brenner, C. Toher & S. Curtarolo, *Settling the matter of the role of vibrations in the stability of high-entropy carbides*, *Nat. Commun.* **12**, 5747 (2021). DOI: [10.1038/s41467-021-25979-5](https://doi.org/10.1038/s41467-021-25979-5). [PDF]
  - This paper was selected for **Editors' Highlight** by Springer Nature (2021).
30. M. D. Hossain, T. Borman, **C. Oses**, M. Esters, C. Toher, L. Feng, A. Kumar, W. G. Fahrenholtz, S. Curtarolo, D. W. Brenner, J. M. LeBeau & J.-P. Maria, *Entropy Landscaping of High-Entropy Carbides*, *Adv. Mater.* **33**(42), 2102904 (2021). DOI: [10.1002/adma.202102904](https://doi.org/10.1002/adma.202102904). [PDF]

29. C. W. Andersen<sup>†</sup>, R. Armiento<sup>†</sup>, E. Blokhin<sup>†</sup>, G. J. Conduit<sup>†</sup>, S. Dwaraknath<sup>†</sup>, M. L. Evans<sup>†</sup>, Á. Fekete<sup>†</sup>, A. Gopakumar<sup>†</sup>, S. Gražulis<sup>†</sup>, A. Merkys<sup>†</sup>, F. Mohamed<sup>†</sup>, **C. Oses**<sup>†</sup>, G. Pizzi<sup>†</sup>, G.-M. Rignanese<sup>†</sup>, M. Scheidgen<sup>†</sup>, L. Talirz<sup>†</sup>, C. Toher<sup>†</sup>, D. Winston<sup>†</sup>, R. Aversa, K. Choudhary, P. Colinet, S. Curtarolo, D. Di Stefano, C. Draxl, S. Er, M. Esters, M. Fornari, M. Giantomassi, M. Govoni, G. Hautier, V. Hegde, M. K. Horton, P. Huck, G. Huhs, J. Hummelshøj, A. Kariryaa, B. Kozinsky, S. Kumbhar, M. Liu, N. Marzari, A. J. Morris, A. Mostofi, K. A. Persson, G. Petretto, T. Purcell, F. Ricci, F. Rose, M. Scheffler, D. Speckhard, M. Uhrin, A. Vaitkus, P. Villars, D. Waroquiers, C. Wolverton, M. Wu & X. Yang, *OPTIMADE: an API for exchanging materials data*, *Sci. Data* **8**, 217 (2021). DOI: [10.1038/s41597-021-00974-z](https://doi.org/10.1038/s41597-021-00974-z). [PDF]
28. R. Friedrich, M. Esters, **C. Oses**, S. Ki, M. J. Brenner, D. Hicks, M. J. Mehl, C. Toher & S. Curtarolo, *Automated coordination corrected enthalpies with AFLOW-CCE*, *Phys. Rev. Mater.* **5**, 043803 (2021). DOI: [10.1103/PhysRevMaterials.5.043803](https://doi.org/10.1103/PhysRevMaterials.5.043803). [PDF]
27. D. Hicks, M. J. Mehl, M. Esters, **C. Oses**, O. Levy, G. L. W. Hart, C. Toher & S. Curtarolo, *The AFLOW Library of Crystallographic Prototypes: Part 3*, *Comput. Mater. Sci.* **199**, 110450 (2021). DOI: [10.1016/j.commatsci.2021.110450](https://doi.org/10.1016/j.commatsci.2021.110450).
26. M. J. Mehl, M. Ronquillo, D. Hicks, M. Esters, **C. Oses**, R. Friedrich, A. Smolyanyuk, E. Gossett, D. Finkenstadt & S. Curtarolo, *Tin-pest problem as a test of density functionals using high-throughput calculations*, *Phys. Rev. Mater.* **5**, 083608 (2021). DOI: [10.1103/PhysRevMaterials.5.083608](https://doi.org/10.1103/PhysRevMaterials.5.083608). [PDF]
25. M. D. Hossain<sup>†</sup>, T. Borman<sup>†</sup>, A. Kumar, X. Chen, A. Khosravani, S. R. Kalidindi, E. A. Paisley, M. Esters, **C. Oses**, C. Toher, S. Curtarolo, J. M. LeBeau, D. W. Brenner & J.-P. Maria, *Carbon Stoichiometry and Mechanical Properties of High Entropy Carbides*, *Acta Mater.* **215**, 117051 (2021). DOI: [10.1016/j.actamat.2021.117051](https://doi.org/10.1016/j.actamat.2021.117051). [PDF]

## 2020

24. A. G. Kusne<sup>†</sup>, H. Yu<sup>†</sup>, C. Wu, H. Zhang, J. Hatrick-Simpers, B. DeCost, S. Sarker, **C. Oses**, C. Toher, S. Curtarolo, A. V. Davydov, R. Agarwal, L. A. Bendersky, M. Li, A. Mehta & I. Takeuchi, *On-the-fly Closed-loop Autonomous Materials Discovery via Bayesian Active Learning*, *Nat. Commun.* **11**, 5966 (2020). DOI: [10.1038/s41467-020-19597-w](https://doi.org/10.1038/s41467-020-19597-w). [PDF]
23. K. Kaufmann, D. Maryanovsky, W. M. Mellor, C. Zhu, A. S. Rosengarten, T. J. Harrington, **C. Oses**, C. Toher, S. Curtarolo & K. S. Vecchio, *Discovery of novel high-entropy ceramics via machine learning*, *npj Comput. Mater.* **6**, 42 (2020). DOI: [10.1038/s41524-020-0317-6](https://doi.org/10.1038/s41524-020-0317-6). [PDF]
22. **C. Oses**, C. Toher & S. Curtarolo, *High-entropy ceramics*, *Nat. Rev. Mater.* **5**, 295–309 (2020). DOI: [10.1038/s41578-019-0170-8](https://doi.org/10.1038/s41578-019-0170-8). [PDF]
  - This paper was highlighted as a “hot paper” by Web of Science (Clarivate Analytics) (Nov 16, 2021).

## 2019

21. D. C. Ford, D. Hicks, **C. Oses**, C. Toher & S. Curtarolo, *Metallic glasses for biodegradable implants*, *Acta Mater.* **176**, 297–305 (2019). DOI: [10.1016/j.actamat.2019.07.008](https://doi.org/10.1016/j.actamat.2019.07.008). [PDF]
20. P. Avery, X. Wang, **C. Oses**, E. Gossett, D. M. Proserpio, C. Toher, S. Curtarolo & E. Zurek, *Predicting Superhard Materials via a Machine Learning Informed Evolutionary Structure Search*, *npj Comput. Mater.* **5**, 89 (2019). DOI: [10.1038/s41524-019-0226-8](https://doi.org/10.1038/s41524-019-0226-8). [PDF]
19. C. Toher, **C. Oses**, D. Hicks & S. Curtarolo, *Unavoidable disorder and entropy in multi-component systems*, *npj Comput. Mater.* **5**, 69 (2019). DOI: [10.1038/s41524-019-0206-z](https://doi.org/10.1038/s41524-019-0206-z). [PDF]
18. R. Friedrich, D. Usanmaz, **C. Oses**, A. R. Supka, M. Fornari, M. Buongiorno Nardelli, C. Toher & S. Curtarolo, *Coordination corrected ab initio formation enthalpies*, *npj Comput. Mater.* **5**, 59 (2019). DOI: [10.1038/s41524-019-0192-1](https://doi.org/10.1038/s41524-019-0192-1). [PDF]
17. P. Nath, D. Usanmaz, D. Hicks, **C. Oses**, M. Fornari, M. Buongiorno Nardelli, C. Toher & S. Curtarolo, *AFLOW-QHA3P: Robust and automated method to compute thermodynamic properties of solids*, *Phys. Rev. Mater.* **3**, 073801 (2019). DOI: [10.1103/PhysRevMaterials.3.073801](https://doi.org/10.1103/PhysRevMaterials.3.073801). [PDF]

## 2018

16. **C. Oses**, E. Gossett, D. Hicks, F. Rose, M. J. Mehl, E. Perim, I. Takeuchi, S. Sanvito, M. Scheffler, Y. Lederer, O. Levy, C. Toher & S. Curtarolo, *AFLOW-CHULL: Cloud-oriented platform for autonomous phase stability analysis*, *J. Chem. Inf. Model.* **58**(12), 2477–2490 (2018). DOI: [10.1021/acs.jcim.8b00393](https://doi.org/10.1021/acs.jcim.8b00393). [PDF]
15. **C. Oses**, C. Toher & S. Curtarolo, *Data-driven design of inorganic materials with the Automatic Flow Framework for Materials Discovery*, *MRS Bull.* **43**(9), 670–675 (2018). DOI: [10.1557/mrs.2018.207](https://doi.org/10.1557/mrs.2018.207). [PDF]
14. P. Sarker<sup>†</sup>, T. J. Harrington<sup>†</sup>, C. Toher, **C. Oses**, M. Samiee, J.-P. Maria, D. W. Brenner, K. S. Vecchio & S. Curtarolo, *High-entropy high-hardness metal carbides discovered by entropy descriptors*, *Nat. Commun.* **9**, 4980 (2018). DOI: [10.1038/s41467-018-07160-7](https://doi.org/10.1038/s41467-018-07160-7). [PDF]
13. V. Stanev, **C. Oses**, A. G. Kusne, E. Rodriguez, J. Paglione, S. Curtarolo & I. Takeuchi, *Machine learning modeling of superconducting critical temperature*, *npj Comput. Mater.* **4**, 29 (2018). DOI: [10.1038/s41524-018-0085-8](https://doi.org/10.1038/s41524-018-0085-8). [PDF]
12. E. Gossett, C. Toher, **C. Oses**, O. Isayev, F. Legrain, F. Rose, E. Zurek, J. Carrete, N. Mingo, A. Tropsha & S. Curtarolo, *AFLOW-ML: A RESTful API for machine-learning prediction of materials properties*, *Comput. Mater. Sci.* **152**, 134–145 (2018). DOI: [10.1016/j.commatsci.2018.03.075](https://doi.org/10.1016/j.commatsci.2018.03.075). [PDF]
  - This paper was selected for Editor’s Choice by Elsevier (2018).
11. D. Hicks, **C. Oses**, E. Gossett, G. Gomez, R. H. Taylor, C. Toher, M. J. Mehl, O. Levy & S. Curtarolo, *AFLOW-SYM: platform for the complete, automatic and self-consistent symmetry analysis of crystals*, *Acta Cryst. A* **74**, 184–203 (2018). DOI: [10.1107/S2053273318003066](https://doi.org/10.1107/S2053273318003066). [PDF]

## 2017

10. A. Hever, **C. Oses**, S. Curtarolo, O. Levy & A. Natan, *The structure and composition statistics of 6A binary and ternary structures*, *Inorg. Chem.* **57**(2), 653–667 (2017). DOI: [10.1021/acs.inorgchem.7b02462](https://doi.org/10.1021/acs.inorgchem.7b02462). [PDF]

9. F. Rose, C. Toher, E. Gossett, **C. Oses**, M. Buongiorno Nardelli, M. Fornari & S. Curtarolo, *AFLUX: The LUX materials search API for the AFLOW data repositories*, *Comput. Mater. Sci.* **137**, 362–370 (2017). DOI: [10.1016/j.commatsci.2017.04.036](https://doi.org/10.1016/j.commatsci.2017.04.036). [PDF]
  - This paper was selected for **Editor's Choice** by Elsevier (2017).
8. O. Isayev<sup>†</sup>, **C. Oses**<sup>†</sup>, C. Toher, E. Gossett, S. Curtarolo & A. Tropsha, *Universal Fragment Descriptors for Predicting Properties of Inorganic Crystals*, *Nat. Commun.* **8**, 15679 (2017). DOI: [10.1038/ncomms15679](https://doi.org/10.1038/ncomms15679). [PDF]
7. C. Toher, **C. Oses**, J. J. Plata, D. Hicks, F. Rose, O. Levy, M. de Jong, M. Asta, M. Fornari, M. Buongiorno Nardelli & S. Curtarolo, *Combining the AFLOW GIBBS and elastic libraries to efficiently and robustly screening thermomechanical properties of solids*, *Phys. Rev. Mater.* **1**, 015401 (2017). DOI: [10.1103/PhysRevMaterials.1.015401](https://doi.org/10.1103/PhysRevMaterials.1.015401). [PDF]
6. C. Nyshadham, **C. Oses**, J. E. Hansen, I. Takeuchi, S. Curtarolo & G. L. W. Hart, *A Computational High-Throughput Search for New Ternary Superalloys*, *Acta Mater.* **122**, 438–447 (2017). DOI: [10.1016/j.actamat.2016.09.017](https://doi.org/10.1016/j.actamat.2016.09.017). [PDF]
5. S. Sanvito, **C. Oses**, J. Xue, A. Tiwari, M. Žic, T. Archer, P. Tozman, M. Venkatesan, J. M. D. Coey & S. Curtarolo, *Accelerated Discovery of New Magnets in the Heusler Alloy Family*, *Sci. Adv.* **3**(4), e1602241 (2017). DOI: [10.1126/sciadv.1602241](https://doi.org/10.1126/sciadv.1602241). [PDF]

2016

4. A. van Roekeghem, J. Carrete, **C. Oses**, S. Curtarolo & N. Mingo, *High-Throughput Computation of Thermal Conductivity of High-Temperature Solid Phases: The Case of Oxide and Fluoride Perovskites*, *Phys. Rev. X* **6**(4), 041061 (2016). DOI: [10.1103/PhysRevX.6.041061](https://doi.org/10.1103/PhysRevX.6.041061). [PDF]
3. K. Yang, **C. Oses** & S. Curtarolo, *Modeling Off-Stoichiometry Materials with a High-Throughput Ab-Initio Approach*, *Chem. Mater.* **28**(18), 6484–6492 (2016). DOI: [10.1021/acs.chemmater.6b01449](https://doi.org/10.1021/acs.chemmater.6b01449). [PDF]

2015

2. C. E. Calderon, J. J. Plata, C. Toher, **C. Oses**, O. Levy, M. Fornari, A. Natan, M. J. Mehl, G. L. W. Hart, M. Buongiorno Nardelli & S. Curtarolo, *The AFLOW Standard for High-Throughput Materials Science Calculations*, *Comput. Mater. Sci.* **108A**, 233–238 (2015). DOI: [10.1016/j.commatsci.2015.07.019](https://doi.org/10.1016/j.commatsci.2015.07.019). [PDF]
  - This paper was selected for **Editor's Choice** by Elsevier (2015).
1. O. Isayev, D. Fourches, E. N. Muratov, **C. Oses**, K. M. Rasch, A. Tropsha & S. Curtarolo, *Materials Cartography: Representing and Mining Materials Space Using Structural and Electronic Fingerprints*, *Chem. Mater.* **27**(3), 735–743 (2015). DOI: [10.1021/cm503507h](https://doi.org/10.1021/cm503507h). [PDF]
  - This paper was one of the **top 10 most highly downloaded papers** for the month of January 2015 by the American Chemical Society (2015).
  - This paper was selected for **Editors' Choice** by the American Chemical Society (2015).

## Book Publications

2019

3. Toher, **C. Oses** & S. Curtarolo, *Automated computation of materials properties*, *Materials Informatics: Methods, Tools and Applications*, Ch. 7. DOI: [10.1002/9783527802265.ch7](https://doi.org/10.1002/9783527802265.ch7). [PDF]

2018

2. S. Sanvito, M. Žic, J. Nelson, T. Archer, **C. Oses** & S. Curtarolo, *Machine learning and high-throughput approaches to magnetism*, *Handbook of Materials Modeling. Volume 2 Applications: Current and Emerging Materials*. DOI: [10.1007/978-3-319-50257-1\\_108-1](https://doi.org/10.1007/978-3-319-50257-1_108-1). [PDF]
1. C. Toher, **C. Oses**, D. Hicks, E. Gossett, F. Rose, P. Nath, D. Usanmaz, D. C. Ford, E. Perim, C. E. Calderon, J. J. Plata, Y. Lederer, M. Jahnátek, W. Setyawan, S. Wang, J. Xue, K. M. Rasch, R. V. Chepulskii, R. H. Taylor, G. Gomez, H. Shi, A. R. Supka, R. Al Rahal Al Orabi, P. Gopal, F. T. Cerasoli, L. Liyanage, H. Wang, I. Siloi, L. A. Agapito, C. Nyshadham, G. L. W. Hart, J. Carrete, F. Legrain, N. Mingo, E. Zurek, O. Isayev, A. Tropsha, S. Sanvito, R. M. Hanson, I. Takeuchi, M. J. Mehl, A. N. Kolmogorov, K. Yang, P. D'Amico, A. Calzolari, M. Costa, R. De Gennaro, M. Buongiorno Nardelli, M. Fornari, O. Levy & S. Curtarolo, *The AFLOW Fleet for Materials Discovery*, *Handbook of Materials Modeling. Volume 1 Methods: Theory and Modeling*. DOI: [10.1007/978-3-319-42913-7\\_63-2](https://doi.org/10.1007/978-3-319-42913-7_63-2). [PDF]

## Teaching Experience

|               |                     |                                                                                                                                                  |
|---------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Instructor    | Spring<br>2023–2026 | EN.500.113: <i>Gateway Computing: Python</i> , Johns Hopkins University                                                                          |
| Instructor    | Fall 2023–2025      | EN.510.666: <i>Introduction to Computational Materials Modeling</i> , Johns Hopkins University                                                   |
| Co-Instructor | Spring 2021         | ME 555: <i>Applications of Artificial Intelligence in Materials</i> , Duke University Department of Mechanical Engineering and Materials Science |

|                    |                       |                                                                                                                                                          |
|--------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Teaching Assistant | Spring 2020           | ME 555: <i>Computational Materials Science by Examples and Applications</i> , Duke University Department of Mechanical Engineering and Materials Science |
| Teaching Assistant | Fall 2014–Spring 2015 | ME 221: <i>Structure and Properties of Solids</i> , Duke University Department of Mechanical Engineering and Materials Science                           |
|                    |                       | • <b>Best Teaching Assistant Award</b> , Aug 14, 2015                                                                                                    |

### Service<sup>†</sup> contributed equally

*Mini Symposium on “Computational Thermodynamics: Energy and Energy Landscape”*

**Co-Organizers:** Z.-K. Liu, J. Deng, T. R. Sinno & R. Wentzcovitch

23. **Co-Organizer and Presenter** at the 2025 SIAM New York-New Jersey-Pennsylvania Section Conference, University Park, PA — Oct 31–Nov 2, 2025.

*Johns Hopkins Data Science and AI Institute Fall 2025 Symposium: AI in Daily Life*

**Co-Organizers:** D. J. H. Mathews<sup>†</sup>, S. Vedula<sup>†</sup>, C. Douville<sup>†</sup>, T. Shu<sup>†</sup>, K. Vaught<sup>†</sup>, B. Ranno, S. Pagano & S. McGhee

22. **Organizer** for the Fall 2025 Symposium at the Johns Hopkins Data Science and AI Institute, Baltimore, MD — Sep 16, 2025.

*Data-Driven Materials Modeling Workshop*

**Co-Organizers:** B. Bukowski & T. Curk

21. **Organizer and Presenter** at Johns Hopkins University, Baltimore, Maryland — May 29–31, 2024.

- *Data-Driven Thermodynamic Modeling for Materials Discovery* recording: <https://youtu.be/kZj3zQkBAKg>

*Foundations to Futures: Materials Data and AI*

**Co-Chairs:** D. Audus & F. Sen

20. **Conference Co-Chair** at the Materials Research Data Alliance (MaRDA) 2024 Annual Meeting, Baltimore, Maryland — Feb 20–22, 2024.

*Focus Session: Computational Design, Understanding and Discovery of Novel Materials*

**Co-Chairs:** E. Jankowski, R. Sundararaman & D. Usanmaz

19. **Session Chair** for the March Meeting of the American Physical Society, Minneapolis, Minnesota — Mar 3–8, 2024.

*AI, Data Science — Developing the Role for Sustainable Energy in Hopkins’ Expansion and Vision*

**Co-Chair:** P. Clancy

18. **Session Co-Chair** at the ROSEI 2024 Summit, Baltimore, Maryland — Jan 17, 2024.



*AFLOW School: Integrated infrastructure for computational materials discovery*

**Co-Organizers:** C. Toher, D. Hicks, M. Esters, R. Friedrich, E. Gossett, A. Smolyanyuk, H. Eckert, S. Divilov, F. Rose, M. J. Brenner & S. Curtarolo

17. **Presenter** for the Machine Learning for Materials Research Bootcamp of the University of Maryland/NIST/MRS, College Park, Maryland — Aug 10, 2023.
16. **Organizer and Presenter** at Johns Hopkins University, Baltimore, Maryland — Sep 21, 2022.
  - *Introduction and AFLOW-ML: Machine Learning* recording: <https://youtu.be/Xj5BGuFC9ew>
15. **Presenter** for the Machine Learning for Materials Research Bootcamp of the University of Maryland/NIST/MRS, College Park, Maryland — Aug 11, 2022.
14. **Co-Organizer and Presenter** at the East African Institute for Fundamental Research, University of Rwanda, Kigali, Rwanda — Feb 21–24, 2022.
13. **Co-Organizer and Presenter** at the Technische Universität (TU) Dresden and Helmholtz-Zentrum Dresden-Rossendorf — Sep 6–10, 2021.
  - *Introduction to Density Functional Theory and VASP* recording: [https://youtu.be/\\_RsQH3TY7kl](https://youtu.be/_RsQH3TY7kl)
  - *AFLOW-CHULL: Thermodynamics* recording: <https://youtu.be/zcY7gTZIB-Y>
  - *AFLOW-POCC: Disorder* recording: <https://youtu.be/lcDSYiF4AS4>
12. **Co-Organizer and Presenter** at the University of Virginia, Charlottesville, Virginia — Aug 17, 2021.
  - *AFLOW-CHULL and AFLOW-CCE: Thermodynamics* recording: <https://youtu.be/cLhOcN1sQ7M>
11. **Presenter** for the Machine Learning for Materials Research Bootcamp of the University of Maryland/NIST, College Park, Maryland — Jul 29, 2021.
  - *AFLOW-ML: Machine Learning* recording: <https://youtu.be/uFQ-lyTaxCc>
10. **Co-Organizer and Presenter** at Texas A&M University, College Station, Texas — Jul 12–15, 2021.
  - *Introduction to Density Functional Theory and VASP* recording: <https://youtu.be/KXnJGdVgosa>
  - *AFLOW-CHULL and AFLOW-CCE: Thermodynamics* recording: <https://youtu.be/ElaniAcrbhU>
  - *AFLOW-POCC: Disorder* recording: [https://youtu.be/D\\_cfHIlpBiA](https://youtu.be/D_cfHIlpBiA)
9. **Session Chair** for the Virtual Spring Meeting of the Materials Research Society — Apr 17, 2021.
8. **Presenter** for the Materials 4.0 Summer School 2020 at the Dresden Center for Computational Materials Science (DCMS), Technische Universität (TU) Dresden — Aug 18, 2020.
  - *AFLOW-CHULL: Thermodynamics* recording: <https://youtu.be/ncm356YNBVc>
7. **Presenter** for the Machine Learning for Materials Research Bootcamp & Workshop on Machine Learning Microscopy Data of the University of Maryland/NIST, College Park, Maryland — Jul 23, 2020.
  - *AFLOW-ML: Machine Learning* recording: <https://youtu.be/x2qeBtOXues>
6. **Co-Organizer and Presenter** at Texas A&M University, College Station, Texas — Jun 16–18, 2020.
  - *Introduction to Density Functional Theory and VASP* recording: <https://youtu.be/ChySAfo2w7g>
  - *AFLOW-CHULL: Thermodynamics* recording: <https://youtu.be/9Sa8D4inJ5w>
  - *AFLOW-POCC: Disorder* recording: <https://youtu.be/xr-mU-1ShQQ>
5. **Presenter** for the Machine Learning for Materials Research Bootcamp & Workshop on Autonomous Materials Research of the University of Maryland/NIST, College Park, Maryland — Aug 05, 2019.
4. **Co-Organizer and Presenter** at the University of Pennsylvania, Philadelphia, Pennsylvania — May 03, 2019.
3. **Co-Organizer and Presenter** at the North Carolina State University, Raleigh, North Carolina — Mar 12, 2019.
2. **Co-Organizer and Presenter** at Carnegie Mellon University, Pittsburgh, Pennsylvania — Jan 21, 2019.
1. **Presenter** for the Machine Learning for Materials Research Bootcamp & Workshop on Machine Learning Quantum Materials of the University of Maryland/NIST/Moore Foundation, Institute for Bioscience & Biotechnology Research in Gaithersburg, Maryland — Aug 02, 2018.

**Press and News Releases**

|                              |              |                                                                                                                                                                                                                                                                                                                                           |
|------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| JHU Energy<br>Institute News | Feb 05, 2026 | <i>Corey Oses Awarded ARPA-E IGNIITE Funding for AI-Driven Fusion Materials Research</i><br><br><a href="https://energyinstitute.jhu.edu/corey-oses-awarded-arpa-e-igniite-funding-for-ai-driven-fusion-materials-research">energyinstitute.jhu.edu/corey-oses-awarded-arpa-e-igniite-funding-for-ai-driven-fusion-materials-research</a> |
| JHU Energy<br>Institute News | Dec 17, 2025 | <i>The Platinum Problem</i><br>• This press release is featured on <a href="https://engineering.jhu.edu/materials/news/oses-and-team-win-2025-nexus-award">JHU Engineering News</a> .<br><br><a href="https://energyinstitute.jhu.edu/the-platinum-problem">energyinstitute.jhu.edu/the-platinum-problem</a>                              |
| JHU Engineering<br>News      | Jun 16, 2025 | <i>Oses and Team Win 2025 Nexus Award</i><br><br><a href="https://engineering.jhu.edu/materials/news/oses-and-team-win-2025-nexus-award">engineering.jhu.edu/materials/news/oses-and-team-win-2025-nexus-award</a>                                                                                                                        |

|                                                            |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| JHU Energy<br>Institute News                               | Oct 16, 2024 | <i>Corey Oses Earns ISMM Early-Career Investigator Award</i><br>• This press release is featured on <a href="#">JHU Engineering News</a> .<br><a href="https://energyinstitute.jhu.edu/corey-oses-earns-ismm-early-career-investigator-award">energyinstitute.jhu.edu/corey-oses-earns-ismm-early-career-investigator-award</a>                                                                                                                                                                                                                                                                               |
| JHU Energy<br>Institute News                               | Jun 12, 2024 | <i>Team Co-Led by Corey Oses Receives SURPASS Award</i><br>• This press release is featured on <a href="#">JHU Engineering News</a> .<br><a href="https://energyinstitute.jhu.edu/team-co-led-by-corey-oses-receives-surpass-award">energyinstitute.jhu.edu/team-co-led-by-corey-oses-receives-surpass-award</a>                                                                                                                                                                                                                                                                                              |
| JHU Engineering<br>News                                    | Jun 21, 2023 | <i>Unlocking the potential of thermoelectric materials</i><br><a href="https://engineering.jhu.edu/news/unlocking-the-potential-of-thermoelectric-materials">engineering.jhu.edu/news/unlocking-the-potential-of-thermoelectric-materials</a>                                                                                                                                                                                                                                                                                                                                                                 |
| Duke<br>Engineering<br>News                                | Oct 11, 2022 | <i>Heat-Proof Chaotic Carbides Could Revolutionize Aerospace Technology</i><br><a href="https://pratt.duke.edu/about/news/heat-proof-chaotic-carbides-could-revolutionize-aerospace-technology">pratt.duke.edu/about/news/heat-proof-chaotic-carbides-could-revolutionize-aerospace-technology</a>                                                                                                                                                                                                                                                                                                            |
| White House<br>Office of Science<br>& Technology<br>Policy | Nov 18, 2021 | <i>Featured Vignette in the November 2021 Materials Genome Initiative Strategic Plan (page 9)</i><br><a href="https://mgi.gov/sites/default/files/documents/MGI-2021-Strategic-Plan.pdf">mgi.gov/sites/default/files/documents/MGI-2021-Strategic-Plan.pdf</a>                                                                                                                                                                                                                                                                                                                                                |
| University of<br>Buffalo                                   | Sep 2019     | <i>Scientists predict new forms of superhard carbon</i><br>• This press release is featured on <a href="#">Phys.org</a> , <a href="#">ScienceDaily</a> , <a href="#">SciTechDaily</a> , and <a href="#">Tribonet</a> .<br><a href="https://buffalo.edu/ubnow/stories/2019/09/zurek-superhard-carbon.html">buffalo.edu/ubnow/stories/2019/09/zurek-superhard-carbon.html</a>                                                                                                                                                                                                                                   |
| Duke<br>Engineering<br>News                                | Nov 2018     | <i>Disordered Materials Could Be Hardest, Most Heat-Tolerant Ever</i><br>• This press release is featured on <a href="#">AAAS EurekAlert!</a> , <a href="#">Phys.org</a> , <a href="#">ScienceDaily</a> , <a href="#">Science Bulletin</a> , <a href="#">Naaju</a> , <a href="#">NewsBeezer</a> , <a href="#">RemoNews</a> , <a href="#">Tech2</a> , and <a href="#">LongRoom News</a> .<br><a href="https://pratt.duke.edu/about/news/chaotic-carbides">pratt.duke.edu/about/news/chaotic-carbides</a>                                                                                                       |
| MRS Bulletin                                               | Aug 2017     | <i>Universal fragment descriptor predicts materials properties</i><br><a href="https://cambridge.org/core/journals/mrs-bulletin/news/universal-fragment-descriptor-predicts-materials-properties">cambridge.org/core/journals/mrs-bulletin/news/universal-fragment-descriptor-predicts-materials-properties</a>                                                                                                                                                                                                                                                                                               |
| UNC Eshelman<br>School of<br>Pharmacy                      | Jun 2017     | <i>Breakthrough Tool Predicts Properties of Theoretical Materials, Finds New Uses for Current Ones</i><br>• This press release is featured on <a href="#">AAAS EurekAlert!</a> , <a href="#">Phys.org</a> , and <a href="#">ScienceDaily</a> .<br><a href="https://pharmacy.unc.edu/news/2017/06/06/breakthrough-tool-predicts-properties-theoretical-materials-finds-new-uses-current-ones">pharmacy.unc.edu/news/2017/06/06/breakthrough-tool-predicts-properties-theoretical-materials-finds-new-uses-current-ones</a>                                                                                     |
| Duke<br>Engineering<br>News                                | Apr 2017     | <i>Computers Create Recipe for Two New Magnetic Materials</i><br>• This press release is featured on <a href="#">Phys.org</a> , <a href="#">Slashdot</a> , <a href="#">Hacker News</a> , <a href="#">Reddit</a> , <a href="#">Engadget</a> , <a href="#">Engineering.com</a> , <a href="#">Science Alert</a> , <a href="#">Azo Materials</a> , <a href="#">Next Big Future</a> , <a href="#">Futurism</a> , <a href="#">New Atlas</a> , and <a href="#">International Business Times</a> .<br><a href="https://pratt.duke.edu/about/news/predicting-magnets">pratt.duke.edu/about/news/predicting-magnets</a> |
| MRS Bulletin                                               | Apr 2015     | <i>Materials fingerprints identified for informatics</i><br><a href="https://doi.org/10.1557/mrs.2015.76">doi.org/10.1557/mrs.2015.76</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Computational<br>Chemistry<br>Highlights                   | Jan 2015     | <i>Materials Cartography: Representing and Mining Materials Space Using Structural and Electronic Fingerprints</i><br><a href="https://compchemhighlights.org/2015/01/materials-cartography-representing-and.html">compchemhighlights.org/2015/01/materials-cartography-representing-and.html</a>                                                                                                                                                                                                                                                                                                             |
| Duke Research                                              | Jan 2015     | <i>Molecular Tornado</i><br><a href="https://research.duke.edu/molecular-tornado">research.duke.edu/molecular-tornado</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Duke Graduate<br>School                                    | Oct 2014     | <i>Competing for NSF Fellowships: Advice from a Current Fellow</i><br><a href="https://gradschool.duke.edu/professional-development/blog/competing-nsf-fellowships-advice-current-fellow">gradschool.duke.edu/professional-development/blog/competing-nsf-fellowships-advice-current-fellow</a>                                                                                                                                                                                                                                                                                                               |
| ERN Conference<br>2013                                     | Feb 2013     | <i>2013 Oral and Poster Presentation Award Winners</i><br><a href="https://new.emerging-researchers.org/2013-oral-and-poster-presentation-winners">new.emerging-researchers.org/2013-oral-and-poster-presentation-winners</a>                                                                                                                                                                                                                                                                                                                                                                                 |

**Honors and Awards**

|                   |              |                                                                                                                                                                                                                                                                                                                             |
|-------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Award             | 2026         | Early Career Award: Inspiring Generations of New Innovators to Impact Technologies in Energy 2025 (IGNIITE 2025), Advanced Research Projects Agency-Energy (ARPA-E)                                                                                                                                                         |
| Publication Award | 2026         | Most Popular Articles Collection, <a href="#">Publication in Mater. Horiz.</a> , Royal Society of Chemistry: Materials Horizons                                                                                                                                                                                             |
| Award             | 2024         | Early-Career Investigator Award in Materials Modelling, International Society of Materials Modeling                                                                                                                                                                                                                         |
| Publication Award | 2024         | Editors' Highlight, <a href="#">Publication in Nat. Commun.</a> , Springer Nature                                                                                                                                                                                                                                           |
| Award             | 2023         | Reviewer of the Year, 2022, npj Computational Materials                                                                                                                                                                                                                                                                     |
| Publication Award | 2022         | Editor's Choice, <a href="#">Publication in Comput. Mater. Sci.</a> , Elsevier                                                                                                                                                                                                                                              |
| Publication Award | Nov 16, 2021 | "Hot paper", <a href="#">Publication in Nat. Rev. Mater.</a> , Web of Science (Clarivate Analytics)<br><ul style="list-style-type: none"> <li>Published in the past two years and received enough citations in July/August 2021 to place it in the top 0.1% of papers in the academic field of Materials Science</li> </ul> |
| Publication Award | 2021         | Editors' Highlight, <a href="#">Publication in Nat. Commun.</a> , Springer Nature                                                                                                                                                                                                                                           |
| Publication Award | 2018         | Editor's Choice, <a href="#">Publication in Comput. Mater. Sci.</a> , Elsevier                                                                                                                                                                                                                                              |
| Publication Award | 2017         | Editor's Choice, <a href="#">Publication in Comput. Mater. Sci.</a> , Elsevier                                                                                                                                                                                                                                              |
| Award             | Aug 14, 2015 | <a href="#">Best Teaching Assistant Award (ME 221)</a> , Duke University Department of Mechanical Engineering and Materials Science                                                                                                                                                                                         |
| Publication Award | 2015         | Editor's Choice, <a href="#">Publication in Comput. Mater. Sci.</a> , Elsevier                                                                                                                                                                                                                                              |
| Publication Award | 2015         | Top 10 most highly downloaded papers for the month of January 2015, <a href="#">Publication in Chem. Mater.</a> , American Chemical Society                                                                                                                                                                                 |
| Publication Award | 2015         | Editors' Choice, <a href="#">Publication in Chem. Mater.</a> , American Chemical Society                                                                                                                                                                                                                                    |
| Fellowship        | 2013–2016    | Graduate Research Fellowship, National Science Foundation                                                                                                                                                                                                                                                                   |
| Award             | Aug 22, 2013 | Best Presentation Award at the MEMS Departmental Retreat, Duke University Department of Mechanical Engineering and Materials Science                                                                                                                                                                                        |
| Award             | Mar 02, 2013 | First Place in Nanoscience and Physics Research Presentation, NSF / AAAS / EHR Emerging Researchers National Conference                                                                                                                                                                                                     |
| Scholarship       | 2011–2013    | Shell Incentive Fund Scholarship                                                                                                                                                                                                                                                                                            |
| Scholarship       | 2010 & 2011  | Xerox Corporation Scholarship                                                                                                                                                                                                                                                                                               |
| Scholarship       | 2010 & 2011  | Intel Academic Award                                                                                                                                                                                                                                                                                                        |
| Grant             | Jun 18, 2010 | Cornell University Unmanned Air Systems Team awarded \$1,000 grant, AUVSI Student Unmanned Aerial Systems Competition                                                                                                                                                                                                       |
| Scholarship       | Fall 2010    | Dean's Honor List, Cornell University College of Engineering                                                                                                                                                                                                                                                                |
| Scholarship       | 2009–2013    | Meinig Family Cornell National Scholars<br><ul style="list-style-type: none"> <li>Awarded by Peter Meinig (Past Chairman of the Board of Trustees at Cornell University)</li> </ul>                                                                                                                                         |